



Full Stack Projects

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Main Goal:

This roadmap describes three full stack projects I plan to build as I grow toward software engineering internships. Each project adds new challenges: first connecting a frontend, API, and database; then supporting multiple users and their content; and finally designing a collaborative system with more complex permissions and workflows. The plans may evolve as I learn, but each project is intended to become a working, documented application.

Contents

Projects:	Project type:	Key Take-away
Internship Application Tracker	Personal productivity app	Full-stack fundamentals
Campus Marketplace	User marketplace	Relationships, image storage, search, authorization

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01 Internship Application Tracker

Goal

I plan to build a web application that helps people organize internship and job applications in one place. Users will be able to record companies and roles, track application stages, save deadlines and interview notes, and review their progress. This will be my first complete full stack project. I want to understand how a React interface, a backend API, and a database work together, from a form submission in the browser to saved information in PostgreSQL. The project also addresses a real need in my own internship search, so I can build something useful while learning the fundamentals employers expect me to understand.

Experience I plan to gain

- Build pages and reusable components with React and TypeScript
- Manage forms, user input, loading states, and errors
- Create REST endpoints with FastAPI and connect them to a frontend
- Model and query relational data with PostgreSQL and SQLAlchemy
- Implement CRUD, validation, authentication, and user-specific access
- Use Git and GitHub, test key user flows, and deploy a complete application

Planned technology

Core stack React · TypeScript · Tailwind CSS · React Router · Python · FastAPI · PostgreSQL · SQLAlchemy · Git/GitHub

Tools and hosting VS Code · Vercel (frontend) · Render (API) · Neon (database)

First release features

- Register, sign in, and manage a personal account
- Create, edit, view, and delete applications
- Track stages such as Interested, Applied, Interview, Offer, and Rejected
- Save application dates, deadlines, and notes
- Search and filter applications; view simple progress statistics

Build roadmap

- 01** Define the first release and sketch its screens
- 02** Build the React interface with sample data and forms
- 03** Create and test FastAPI endpoints
- 04** Connect PostgreSQL and replace sample data with database records
- 05** Add authentication, validation, search, and user-specific access
- 06** Test the key flows, document setup, and deploy

02 Campus Marketplace

Goal

I plan to build a marketplace where students can post items for sale, browse what other students have listed, search by keyword or category, and save listings they may want to revisit. Sellers will be able to manage their own posts and include photos and item details.

This project will build on the full stack foundation of the tracker and introduce shared data and user ownership. I want to learn how applications support many users safely: how records relate to one another, how image uploads work, and how the backend checks that a person can only change their own listing. Those patterns are useful across marketplaces, social platforms, booking products, and other real applications.

Experience I plan to gain

- Design one-to-many and many-to-many database relationships
- Build reusable listing cards, filters, and responsive pages
- Add image upload and external file storage
- Implement search, category filters, and saved favorites
- Enforce authorization and ownership in backend routes
- Design APIs for users, listings, and favorites

Planned technology

Core stack React · TypeScript · Tailwind CSS · React Router · Python · FastAPI · PostgreSQL · SQLAlchemy

Tools and hosting Git/GitHub · Swagger UI or Postman · Cloudinary or equivalent image storage · Vercel · Render · Neon

First release features

- Create an account and profile
- Create, edit, and remove personal listings
- Add title, description, price, category, and photos
- Browse, search, and filter listings
- Save and remove favorite listings
- View seller information while protecting listing ownership

Build roadmap

- 01** Plan the first release and map users, listings, and favorites
- 02** Build the marketplace interface with sample listings
- 03** Design the database and create listing APIs
- 04** Connect React, FastAPI, and PostgreSQL; add accounts and permissions
- 05** Add image storage, categories, search, and favorites
- 06** Test browsing and ownership rules, document setup, and deploy

What I want to be able to show

By completing these projects, I aim to demonstrate steady growth: building and deploying a complete application, handling real multi-user data responsibly, and making thoughtful choices in a collaborative system. I also want each project to include a clear README, screenshots, setup instructions, and an honest account of what I learned and what I would improve next.